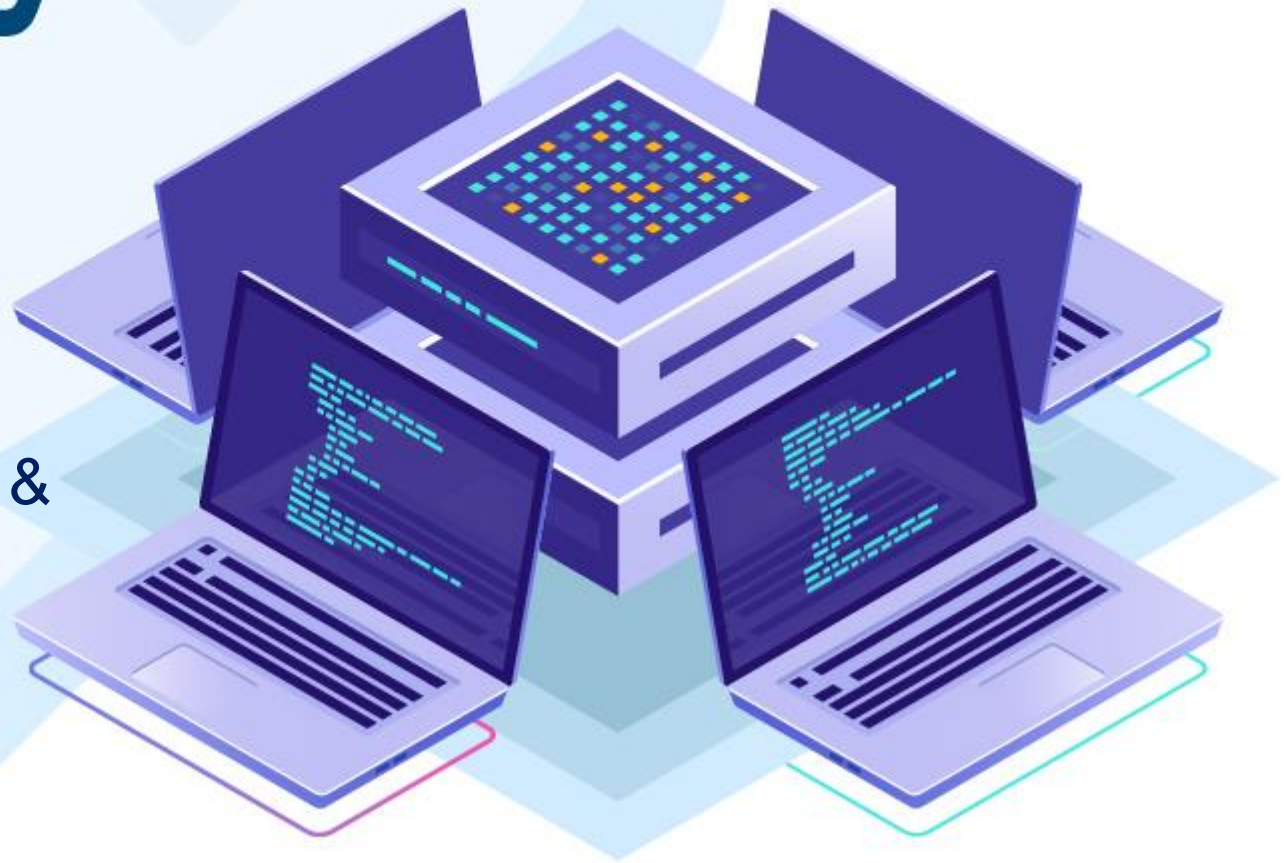
Information Technology

Year: 2026-2027

Fall Semester



- Computer Graphics
- **Dr. Yasmine Mahmoud**

Introduction to Computer Graphics &
Overview of Graphics systems

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الجامعة المصرية للتعلم الإلكتروني الأهلية

THE EGYPTIAN E-LEARNING UNIVERSITY

Student assessments + Assessment Grades

10 Project

10 Quizzes (2 quizzes each **5**) - FTF/ Lectures

5 Assignment + **5** Section attendance

20 Midterm Exam

50 Final Exam

1. What is Computer Graphics?

- Computer graphics is the field of computer science that deals with generating (creating), manipulating (Editing, moving), and displaying (representing) visual content using computers.
- It includes everything from simple 2D images to complex 3D models, animations, and simulations.
- It deals with creating visual effects, movies, games, and virtual reality systems.

2. Applications of Computer Graphics

- **Computer graphics is used in many different fields:**
  - Movies: Special effects like explosions, computer-generated characters, and animated environments.



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Movies


expression
depth of field
fracture
motion
water
reflection
hair

2. Applications of Computer Graphics

- **Computer graphics is used in many different fields:**
  - Video games: Creating real-time rendered environments, character models, and interactive simulations.



2. Applications of Computer Graphics

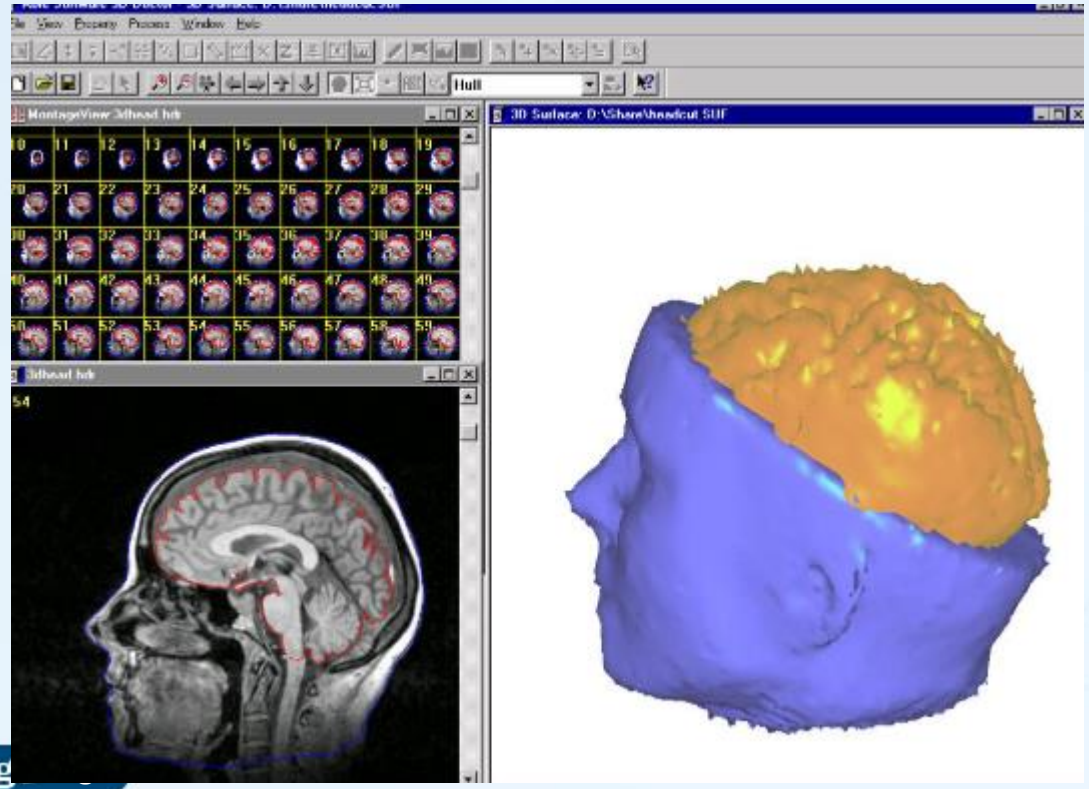
- **Computer graphics is used in many different fields:**
  - Virtual & Augmented Reality: Immersive experiences where users interact with 3D worlds in real-time.

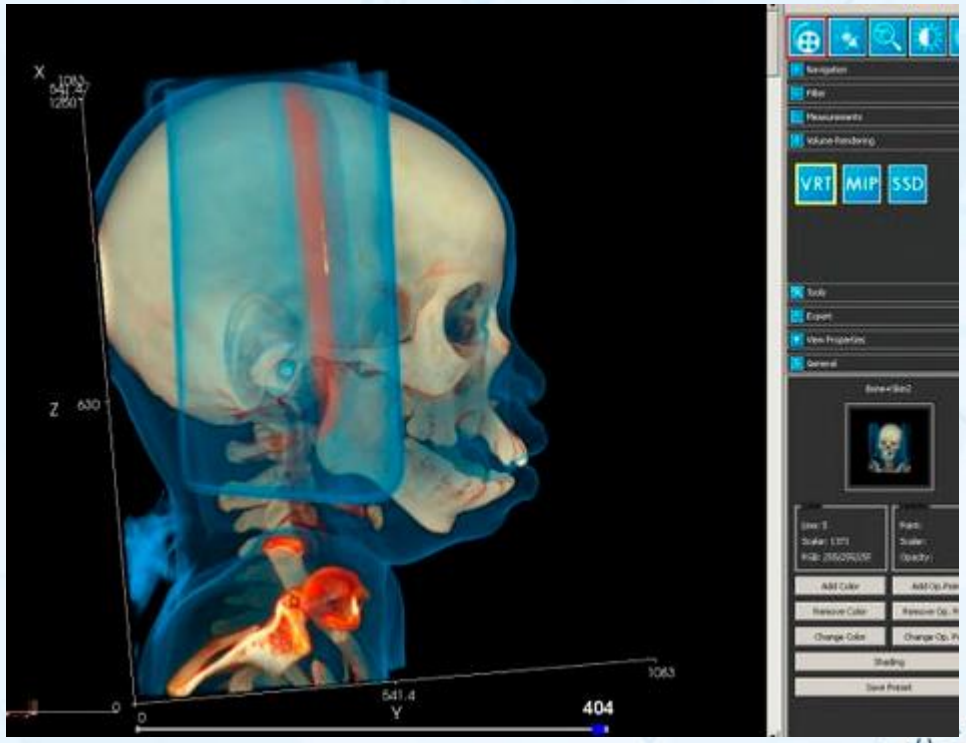




2. Applications of Computer Graphics

- **Computer graphics is used in many different fields:**
  - Medical imaging: 3D reconstructions of CT scans or MRI scans.





2. Applications of Computer Graphics

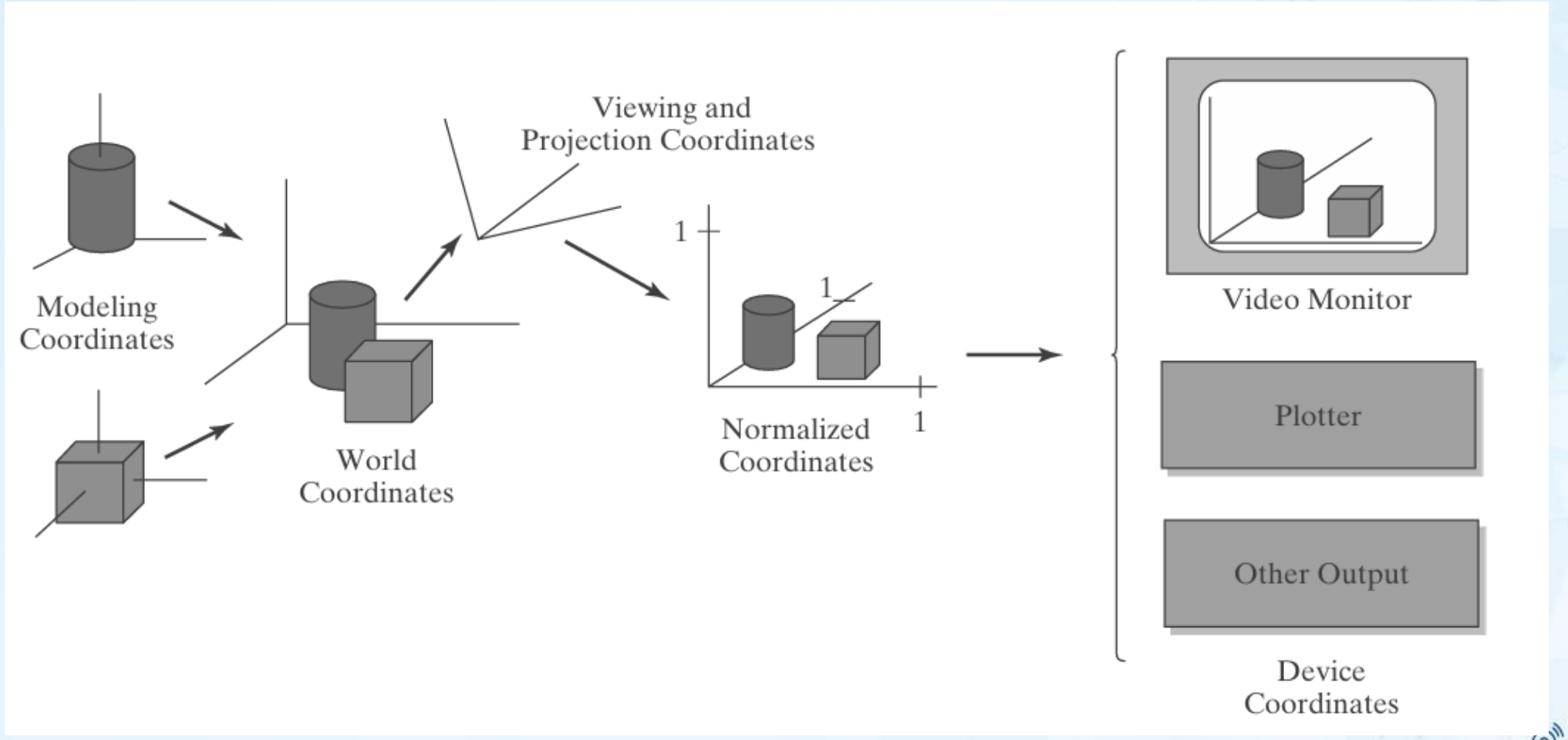
- **Computer graphics is used in many different fields:**
  - Architecture: Visualizing building designs and models through 3D rendering software.



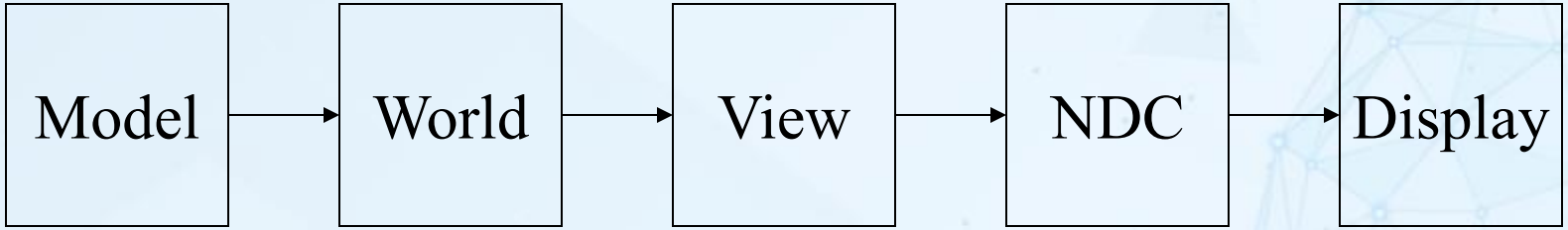
3. The Graphics Pipeline

- **The graphics pipeline** is the sequence of steps involved in turning a description of a scene into a final rendered image displayed on a screen.
- **The primary stages:**
  1. Modeling: Creating 3D models of objects.
  2. Transformation: Manipulating and positioning objects in 3D space.
  3. Lighting: Calculating how light interacts with objects to create realistic shading and highlights.
  4. Rasterization: Converting 3D information into a 2D image (essentially turning models into pixels).
  5. Shading: Determining the color and brightness of pixels based on the lighting and material properties.

From model to image



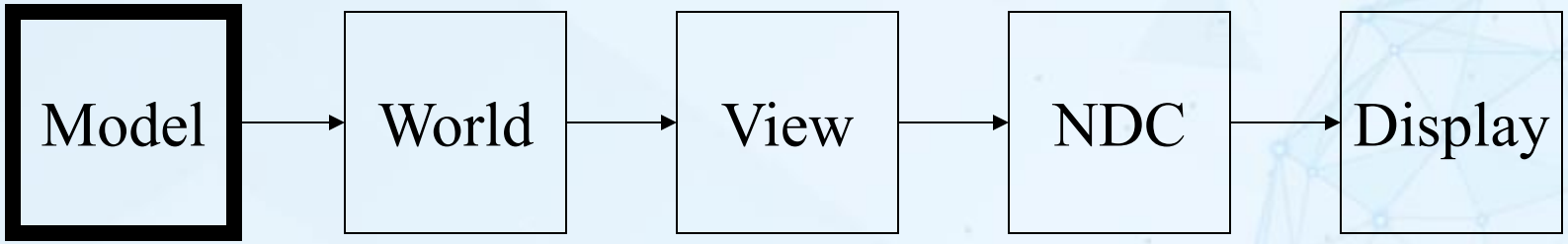
From model to image

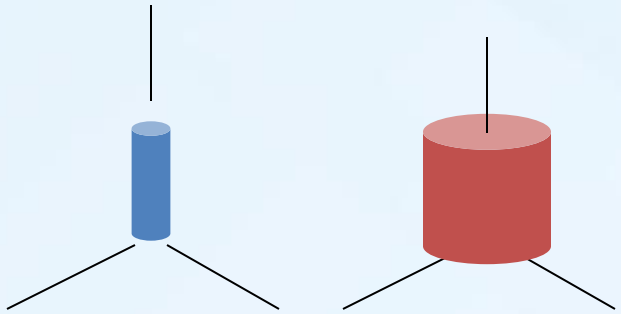


Graphics pipeline

Coordinates and transformations

From model to image





Cylinder:

$$x^2 + y^2 = r^2$$

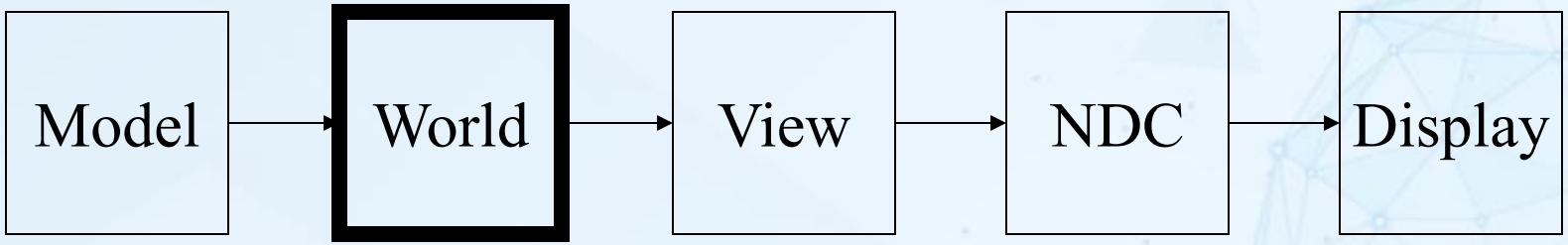
$$0 \le z \le h$$

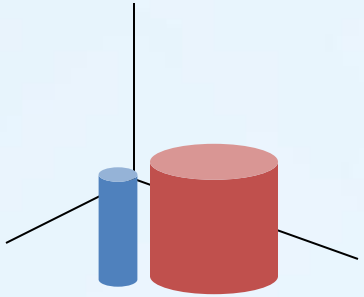
Local or modeling coordinates

Geometric modeling

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From model to image

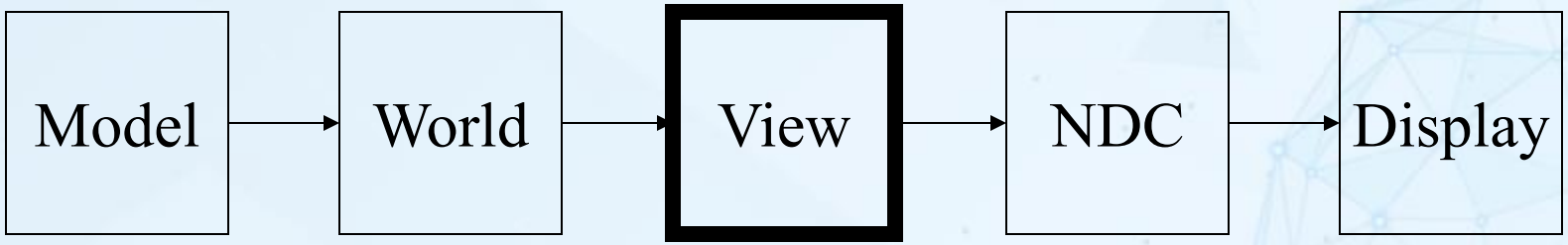


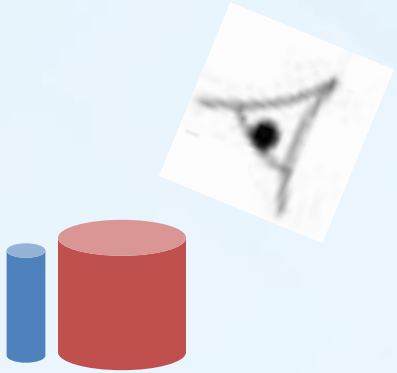


Position cylinders in scene:

World coordinates

From model to image



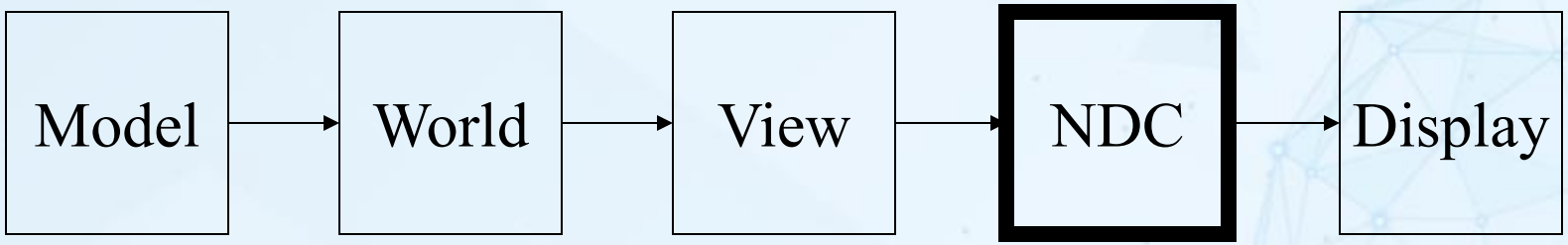


Look at cylinders,
project on virtual screen

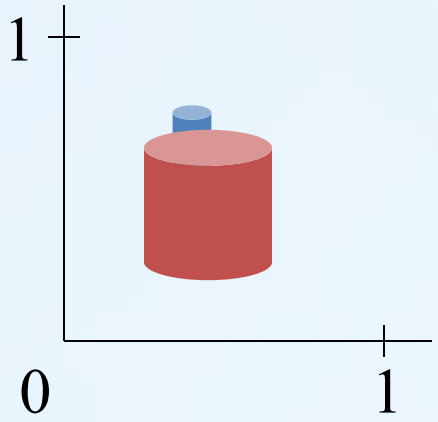
Viewing coordinates

Visible surfaces, shading

From model to image

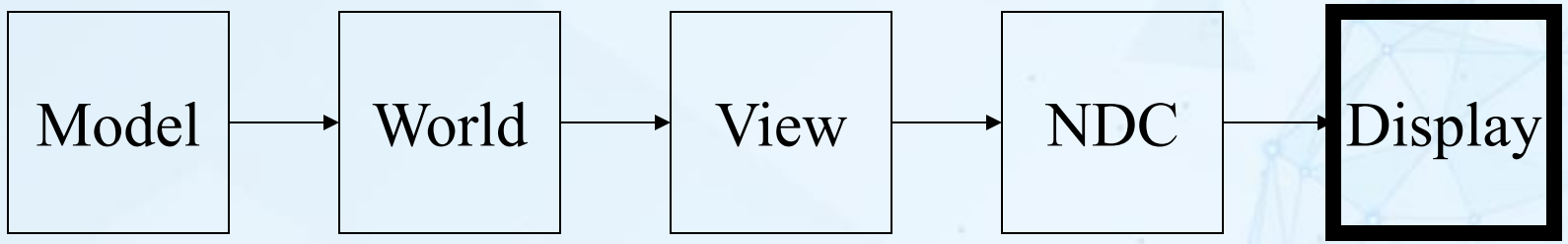


Display:

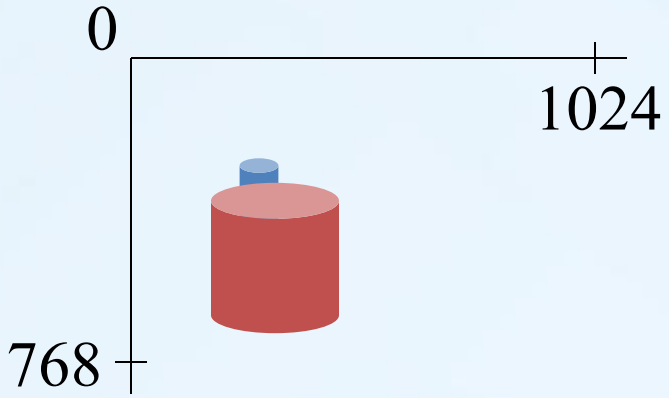


Normalized Device Coordinates

From model to image



Display on screen:



Device Coordinates

Interaction

4. Types of Computer Graphics

- **There are two kinds of computer graphics are:**
  1. Interactive Computer Graphics
  2. Non-Interactive Computer Graphics

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4. Types of Computer Graphics

1. Interactive Computer Graphics

- In interactive computer graphics, users have some controls over the image, the user can make any changes to the image produced.

- Interactive Computer Graphics involves computer-user two-way communication.

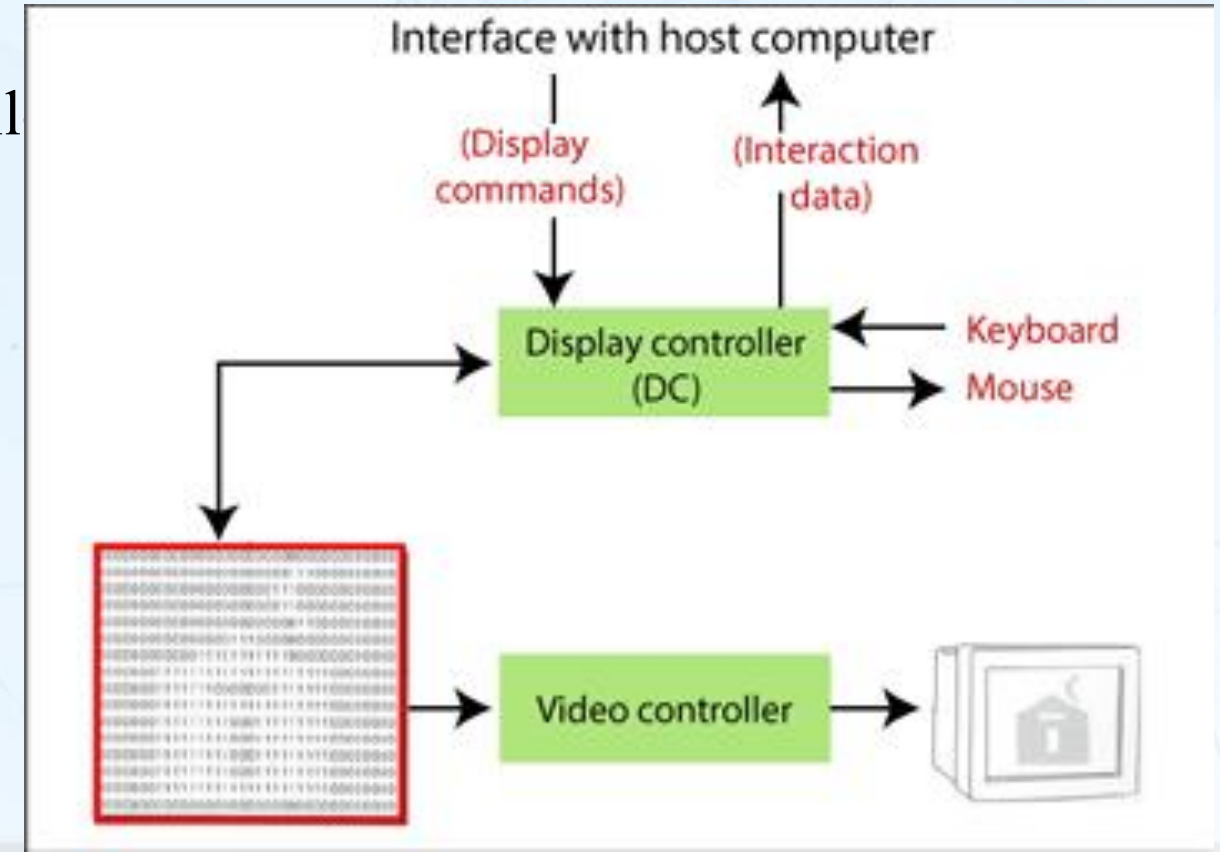
Like:

- Ping-pong game.
- Drawing on touch screens.
- Graphics animation in video games.

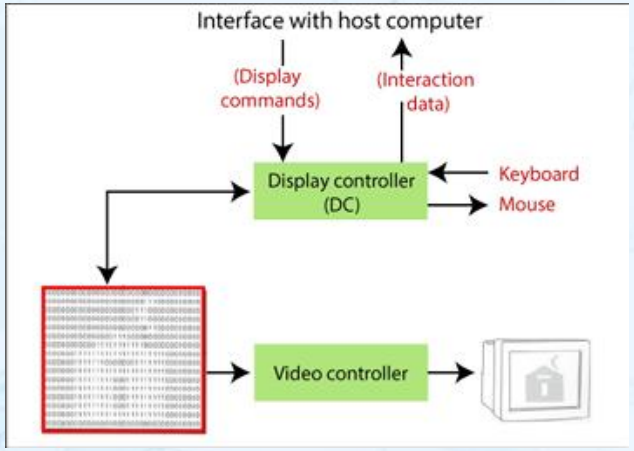
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4. Types of Computer Graphics

- **Working of Interactive Computer Graphics**
  - The modern display of graphics is very simple to build. It is composed of three components:
    A. Display controller or video controll
    B. Digital memory or frame buffer.
    C. Television monitor.



4. Types of Computer Graphics



A. Display controller or video controller:

- It's a Memory Buffer and TV Monitor interface. Its task is to pass Frame Buffer's contents to the monitor.
- The display controller reads each continuous byte of Memory frame buffer data and converts 0's and 1's into appropriate video signals.
- In today's term, the display controller is recognized as a display card, and one of our choices can be a **VGA** (Video Graphics Array) card with a resolution of 640×480.
- Display Controller is also capable of displaying the image in colors.

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4. Types of Computer Graphics

B. Digital memory or frame buffer:

- This is a place where images and pictures are stored as an array (matrix of 0 & 1, 0 represents darkness, and 1 represents image or picture). It is also called a frame buffer.
- In today's term frame buffer is called V-RAM (video RAM), and it helps to store the image in bit form.
- It helps to increase the speed of graphics. Why??

4. Types of Computer Graphics

B. Digital memory or frame buffer:

- This is a place where images and pictures are stored as an array (matrix of 0 & 1, 0 represents darkness, and 1 represents image or picture). It is also called a frame buffer.
- In today's term frame buffer is called V-RAM (video RAM), and it helps to store the image in bit form.
- It helps to increase the speed of graphics. Why???
  - The graphics hardware can quickly read and write to the frame buffer without involving the slower main CPU or RAM

4. Types of Computer Graphics

3. Television monitor:

– Monitor helps us to view the display.

4. Types of Computer Graphics

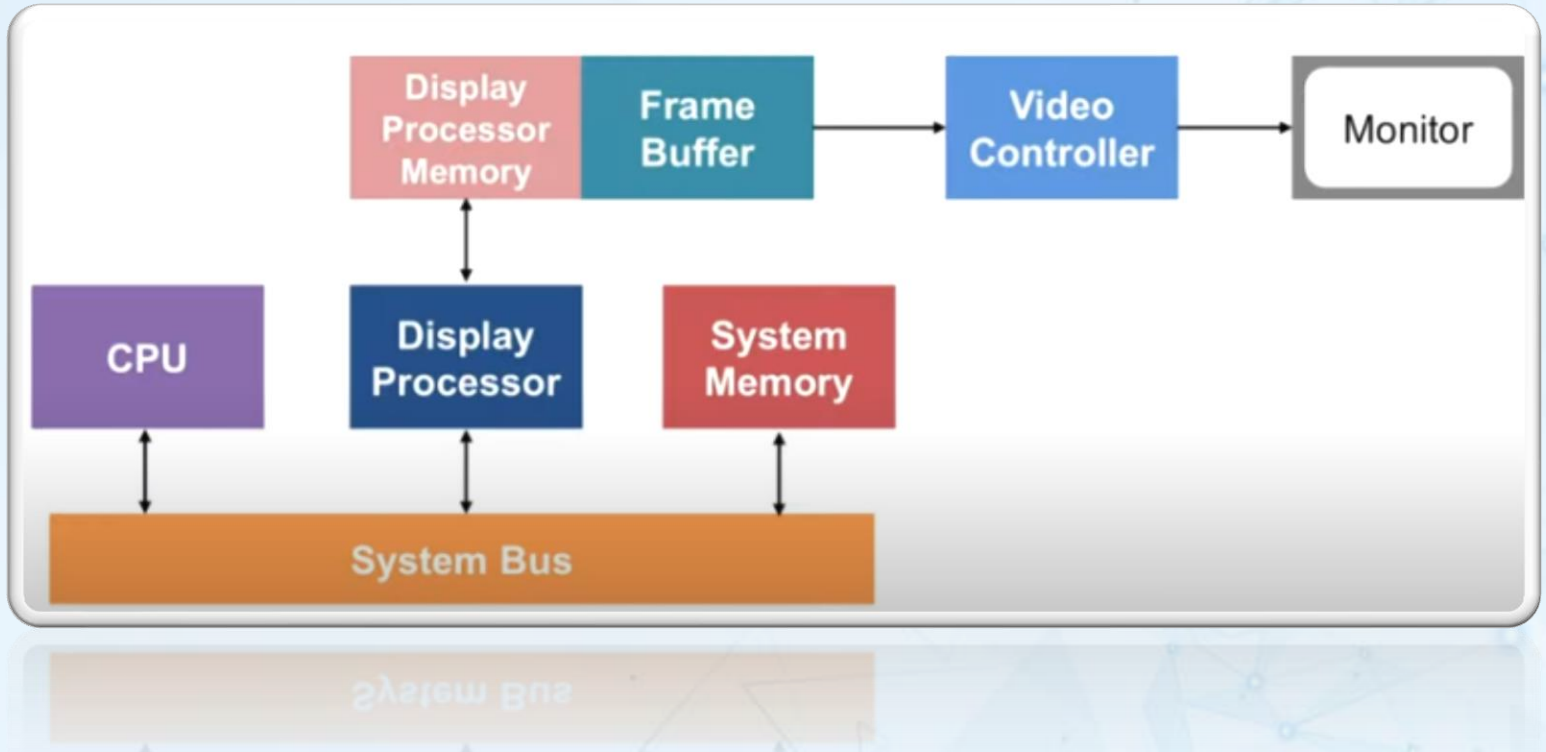
2. Non- Interactive Computer Graphics

- Non-interactive computer graphics are also known as passive computer graphics.
- It is a type of computer graphics in which the user has no control over the image. The photo is completely controlled by the instructions of the program, not by the user.

Like:

- Screen savers.
- Graphic elements are used in the text, document, and PDF presentation.
- Static images are used in mobile applications and websites.
- Business graphics are used as brochures, business cards, menu of the hotel.

Architecture of a Graphics System



Important Terms

- **Rasterization**
  - It is the process of converting vector graphics (descriptions of shapes, lines, and objects) into a raster image (pixels).
  - It is the transformation of a mathematical model of an image into a grid of pixels for display on a screen or output device.
  - Rasterization is often used in real-time rendering like video games.

Important Terms

- **Vector**
  - A vector is a mathematical object that describes a point in space and can represent both position and direction.
  - In graphics, **vector graphics** are used to represent shapes and objects using paths, lines, and curves defined by mathematical equations.
  - These **vector graphics** are resolution-independent, meaning they can be scaled without losing quality.

Important Terms

- **Frame Buffer**
  - A frame buffer is a portion of memory used to store the image data that is going to be displayed on a screen.
  - It acts as a temporary storage that holds the pixel data of an image before it is rendered on a monitor or display.
  - Essentially, it is the memory buffer that contains the color values for each pixel on the screen at a specific time.
  - The **frame buffer** holds the final rendered image **from** the rendering process, and its contents are transferred to the display hardware (e.g., graphics card) to be shown on the screen.

Important Terms

- **Advantages of Frame Buffers**
  - The frame buffer allows for fast and efficient rendering of images by providing a dedicated area in memory for image storage.
  - Using double or triple buffering ensures smooth visual transitions without screen tearing or flickering.
  - Frame buffers can work with different screen resolutions, allowing the same application to render correctly on various display sizes and types.

5. Real-Time vs. Offline Rendering

- **Real-time rendering:**
  - This is used in interactive applications such as video games, where each frame must be rendered quickly (usually aiming for 60 frames per second or higher).
  - Speed is prioritized over quality, and techniques like rasterization are used to render images faster, sometimes sacrificing realism.
- **Offline rendering:**
  - This is used in non-interactive applications such as film production, where high quality is prioritized.
  - Each frame can take minutes or even hours to render.

6. Representation of graphics

- Graphics can be divided into 2D (two-dimensional) and 3D (three-dimensional) graphics.
- In 2D graphics, images are created on a flat plane, while in 3D graphics, images are created with depth, allowing for perspective and volume.

6. Representation of Graphics

- **There are two ways for graphics representation:**
    1. Raster (Bitmap) Graphics
    2. Vector Graphics

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6. Representation of graphics

1. Raster Graphics:

- In raster graphics, the image is presented as a rectangular grid of colored squares.
- Raster images are also called bitmap images. Bitmap images are stored as the collection of small individual dots called pixels.
- Bitmap images require high resolution and anti-aliasing for a smooth appearance.
- Like: Paint, Photoshop, etc.

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6. Representation of graphics

2D Raster Graphics

- The resolution of a 2D raster image refers to the number of pixels in the image.
  - A higher resolution means more pixels, resulting in better detail and sharpness.
  - An image with a resolution of 1920x1080 has 1920 pixels along the width and 1080 pixels along the height.

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6. Representation of graphics

- **2D Raster graphics characteristics:**
  1. Resolution-dependent: Raster images have a fixed resolution, so the quality of the image is tied to its pixel dimensions (e.g., 1920x1080). When you scale up a raster image, it can lose quality and become pixelated or blurry.
  2. Color Information: Each pixel in a raster image has a defined color value, which is typically stored in formats like RGB or CMYK.
  3. Detail and Texture: Raster images are good for complex and detailed images, like photographs, textures.

6. Representation of graphics

2. Vector Graphics

- Vector graphics are images created using mathematical equations and geometric shapes like lines, curves, polygons, and circles. These shapes are defined by vectors that tell the computer how to draw the image.
- Examples: SVG, EPS.

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6. Representation of graphics

- **Vector graphics characteristics:**
  1. Resolution-independent: Unlike raster images, vector graphics are not tied to a specific resolution. You can scale them infinitely without losing quality, as the image is recalculated based on the mathematical equations.
  2. File Size: Vector files are often smaller in size compared to high-resolution raster images **because** they store the mathematical description of shapes, not individual pixels.
  3. Simplicity: Best suited for images with clean lines and simple shapes, such as logos, illustrations, and diagrams.

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THANK YOU FOR WATCHING

QUESTIONS?

